JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 94, NO. B2, PAGES 1827–1842, FEBRUARY 10, 1989

Paleomagnetism of the Oligocene Kalamazoo Tuff: Implications for Middle Tertiary Extension in East Central Nevada

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The Oligocene Kalamazoo Tuff (≈35 Ma) was sampled for paleomagnetic analysis across a 100-km-wide zone of highly extended crust in east central Nevada to estimate between-site vertical axis rotations and thus the relative importance of strike-slip faulting to the mechanism of extension. Subordinate sampling was also undertaken in a younger hornblende dacite intrusion (≈35 Ma) and the overlying tuff of North Creek (≈35 Ma). In general, ash flow tuffs cool and are magnetized within a short period of time relative to geomagnetic secular variation and their eutaxitic structures typically indicate the paleohorizontal. These attributes of ash flow tuffs, in addition to their generally widespread occurrence, make them excellent candidates for combined paleomagnetic and structural studies of regional deformation. However, in this study a number of problems associated with the paleomagnetism of the Kalamazoo Tuff are documented; they include topography-related variations of the eutaxitic structures, secular variation during slow cooling, postblocking rheomorphic flow, and local remagnetization of the tuff by younger igneous activity. The tilt-corrected data, with these sources of error reduced or eliminated, exhibit a 28° ± 12° clockwise rotation of the Schell Creek Range relative to the Kern Mountains region. This rotation implies differential extension accommodated by strike-slip faulting or N-S shortening. The paleomagnetic results also suggest that large changes in strike of layered units near faults with presumed strike-slip movement need not be the result of oroclinal bending, but could result from superimposed sets of orthogonal normal faults.

INTRODUCTION

Exposed in the hills and mountain ranges of east central Nevada are rocks that were involved in an intense episode of crustal extension in middle Tertiary time. The extension is characterized by a subhorizontal detachment fault above which supracrustal rocks have been thinned by brittle deformation and below which midcrustal igneous and metamorphic rocks have been ductilely deformed [e.g., *Coney*, 1979; *Crittenden et al.*, 1980]. Imbricate low-angle faults within the supracrustal rocks place younger strata on older and are interpreted by *Gans* [1982] and *Gans and Miller* [1983] to have formed as high-angle normal faults that were rotated to low dip angles during progressive extension. The zone of WNW-ESE directed extension is 120 km wide from west to east and includes part of the northern Egan, Schell Creek, and Snake Ranges (Figure 1) and is bordered by relatively unextended regions. Both range-by-range palinspastic reconstructions and restoration of older Mesozoic structures indicate that the magnitude of extension averages approximately 250% or 95 km across this zone [*Gans and Miller*, 1983].

The major objective of this study is to assess, using paleomagnetic data, the role of vertical axis rotations within the highly extended region of eastern Nevada. Rotations of fault blocks around near-vertical axes are a likely consequence of crustal extension within domains of strike-slip faults [e.g., *Ron et al.*, 1984]. Although strike-slip faulting is considered subordinate to normal faulting in the overall Cenozoic deformation of the Basin and Range province, paleomagnetic studies have determined block rotations near Lake Mead [*Ron et al.*, 1986] and in west central Nevada [*Hudson and Geissman*, 1987] which imply that locally, regions of extension are associated with strike-slip faults as well as normal faults. In eastern Nevada, north-south variations in the magnitude and direction of tilting and the discontinuous nature of many of the major fault systems suggest that the extensional strain was heterogeneous and that strike-slip faults played a significant role in the extension. Large changes in the strike of layered rocks near faults with apparent strike-slip displacement suggest that in places vertical axis rotations may have been large (as much as 90°). A paleomagnetic study of crustal deformation in the vicinity of the Las Vegas Valley Shear Zone by *Nelson and Jones* [1987] also has documented large rotations of relatively small blocks and has shown that in this region the crust can be modeled as a thin non-Newtonian viscous plate. Similarly, documentation of vertical axis rotations within the highly extended zone of east central Nevada, could assess the role of strike-slip faulting and provide important constraints on the geometry, kinematics, and lateral heterogeneity of extension within the brittle crust.

The tuff unit called the Kalamazoo Tuff by *Gans and Miller* [1983] (equivalent in part to the Kalamazoo Volcanics as used by *Young* [1960]) is the focus of this study because it provides an exceptionally good structural and paleomagnetic marker horizon within the extended region. Erupted relatively early during the episode of middle Tertiary extension, the Kalamazoo Tuff spans the entire extensional zone (Figure 1) and is easily distinguished from other eruptive units in this region by its distinctive compositional and mineralogical zonation, a well-developed eutaxitic structure, and abundant carbonate lithic fragments [*Gans et al.*, 1989]. Single cooling units of ash flow tuffs are usually magnetized

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Paper number 88JB03851.
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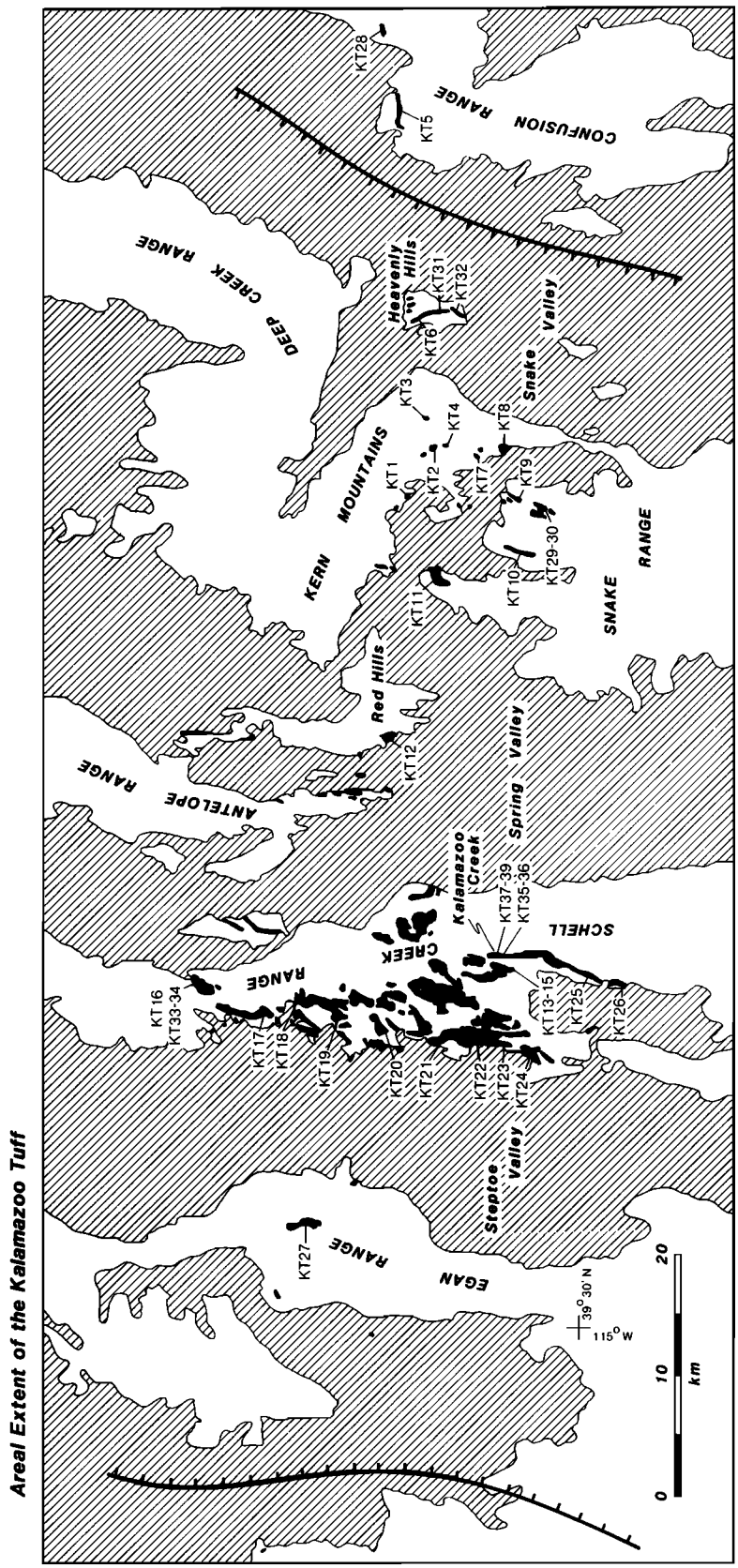


Fig. 1. Regional map of east central Nevada and western Utah showing exposures of the Kalamazoo Tuff of *Gans and Miller* [1983] (black areas) and the paleomagnetic sampling sites (labels) within them. Hachured lines indicate the eastern and western limits of middle Tertiary extension.

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over very short periods of time relative to secular variation of the geomagnetic field and have been correlated over large areas on the basis of their uniform and distinct paleomagnetic directions [e.g., *Cox*, 1971; *Grommé et al.*, 1972; *Best et al.*, 1973; *Nairn et al.*, 1975].

The eutaxitic structure of a welded tuff is generally presumed to indicate the paleohorizontal, and between-site vertical axis rotations can be determined from the site mean paleomagnetic directions of a single ash flow sheet after correction for structural tilt [e.g., *Hillhouse and Wells*, 1986]. The level of success in using this technique depends on both the accuracy of the tuff in recording the ambient geomagnetic field and in the accuracy of the eutaxitic structure in representing the structural tilt. Most ash flow tuffs are excellent paleomagnetic recorders [e.g., *Dalrymple et al.*, 1965; *Grommé et al.*, 1972; *Reynolds*, 1977], but error in determining their paleomagnetic directions can result from incorrect structural corrections due to initial variations in the attitude of the eutaxitic structure [e.g., *Shuey et al.*, 1976; *Geissman et al.*, 1982], secular variation of the geomagnetic field during cooling [e.g., *Hillhouse and Wells*, 1986], rheomorphic flow after blocking of the magnetization [e.g., *Rosenbaum*, 1986; *Geissman*, 1980], and remagnetization by subsequent igneous events as suggested by *Gose* [1971]. In this study, these sources of error have been encountered and for the most part removed from the data set in order to best determine vertical axis rotations of crustal blocks associated with the middle Tertiary extension.

Geologic Setting

Extension in east central Nevada appears to have begun at about 36 Ma, synchronous with the eruption of the earliest volcanic rocks [*Gans et al.*, 1989]. These early volcanic rocks were intermittently erupted from small centers scattered across the region prior to eruption of the voluminous sequence of rocks that includes the Kalamazoo Tuff. K-Ar biotite ages of 34.6 ± 1.2 and 34.0 ± 1.1 Ma [*Gans et al.*, 1989], 36.0 ± 1.4 Ma [*Armstrong*, 1970], and 35.1 ± 2.6 Ma [*McKee et al.*, 1976] have been determined for samples of the Kalamazoo Tuff (all ages are calculated by the new decay constants [*Steiger and Jager*, 1977]), and together yield a weighted mean age of 34.7 ± 0.7 Ma. The Kalamazoo Tuff typically rests on either upper Paleozoic strata or early Oligocene volcanic and sedimentary sequences. The deposition of lower Tertiary rocks exclusively on upper Paleozoic rocks with only minor angular discordance implies that the miogeoclinal stratigraphy at that time was structurally intact and nearly flat-lying [*Gans and Miller*, 1983].

Complete sections of the Kalamazoo Tuff in the central area of exposure are 200–300 m thick and decrease in peripheral areas to less than 100 m thick, indicating, after palinspastic reconstruction, a total magma volume of approximately 240–320 km^3 [*Gans et al.*, 1989]. The vent region for the Kalamazoo Tuff is inferred by *Gans et al.* [1989] to be near Kalamazoo Creek (Figure 1) where some of the thickest sections occur and where the lithic fragments included in the tuff are largest and most abundant. A poorly exposed section of tuff in the Red Hills (Figure 1) may exceed 650 m in thickness and also suggests a nearby source region, perhaps buried beneath the northern end of Spring Valley [*Gans*, 1987].

The Kalamazoo Tuff is zoned from a crystal-poor rhyolite at its base to a crystal-rich hornblende dacite at its top [*Gans*, 1987]. At least four cooling units have been identified and are informally lettered A through D in ascending order. Exposures of units A and D are largely restricted to the upper drainage of Kalamazoo Creek. The most widespread of the four units, unit B, is crystal poor with relatively large pumice fiamme and abundant carbonate lithic fragments. This unit ranges in thickness from over 200 m in the Schell Creek Range to a few meters in distal exposures. Unit B grades upwards from a thin nonwelded base to a partially to densely welded vitrophyre, which is in turn overlain by a thick densely welded devitrified upper portion. Unit C contains more crystals, fewer and smaller lithic fragments, and much smaller fiamme than unit B. Only a partial cooling break separates B from C, and mineralogical and compositional changes appear gradational between these two units [*Gans*, 1987].

Geographically and stratigraphically associated with the Kalmazoo Tuff are crystal-rich hornblende dacite lavas, breccias, and shallow intrusions. They have yielded K-Ar biotite ages of 35.0±1.2 and 34.6±1.2 Ma [*Gans et al.*, 1989] and of 36.1 ± 2.6 and 34.8 ± 2.0 Ma [*McKee et al.*, 1976] for a weighted mean age of 34.9 ± 0.8 Ma. The dacite lavas typically directly overlie the Kalamazoo Tuff and are commonly the youngest eruptive unit exposed in any given area. The massive and monotonous character of the hornblende dacites makes it difficult to estimate their structural attitude or even to distinguish between the lava flows and intrusive rocks. The age, modal mineralogy, and composition of the dacites closely resemble the last eruptive phase of the Kalamazoo Tuff, suggesting that the two are genetically related [*Gans*, 1987]. The lack of a soil horizon or any evidence of erosion at the top of the Kalamazoo Tuff also indicates that the dacites were emplaced soon after the tuff. An especially large hypabyssal intrusion is exposed in the region between the northern Snake Range and the Kern Mountains (Figure 2). The scattered exposures of Kalamazoo Tuff in this area occur as steeply tilted roof pendants within the large intrusion. The large volume and broad areal extent of dacites in east central Nevada suggest that much of the region was underlain by a dacitic magma system of batholithic dimensions in Oligocene time [*Gans et al.*, 1989].

The hornblende dacites are locally overlain by a younger tuff, informally refered to as the tuff of North Creek, which consists of several cooling units. Much smaller in volume and areal extent (Figure 3) than the Kalamazoo Tuff, the modal composition and geochemistry of the tuff of North Creek are also quite different than those of the Kalamazoo Tuff and hornblende dacites. However, this unit has yielded K-Ar biotite ages of 35.9 ± 1.3, 35.5 ± 1.1, 35.2 ± 1.0, and 34.7 ± 1.1 Ma for a weighted mean of 35.2 ± 0.6 Ma [*Gans et al.*, 1989], indicating its eruption closely followed that of the Kalamazoo Tuff and hornblende dacites.

The character of the early Oligocene unconformity and the ages of syntectonic intrusive and volcanic rocks in east central Nevada suggest that most of the faulting and tilting associated with extension occurred in Oligocene time [*Gans and Miller*, 1983]. Originally high-angle normal faults in the Egan and Schell Creek Ranges (Figure 1) apparently penetrated to paleodepths greater than 10 km without flattening significantly; the basal detachment fault is not exposed in this region. However, the ductile-brittle transition is exposed in the northern Snake Range and was initially formed 6–7 km in depth. The overlying supracrustal rocks were

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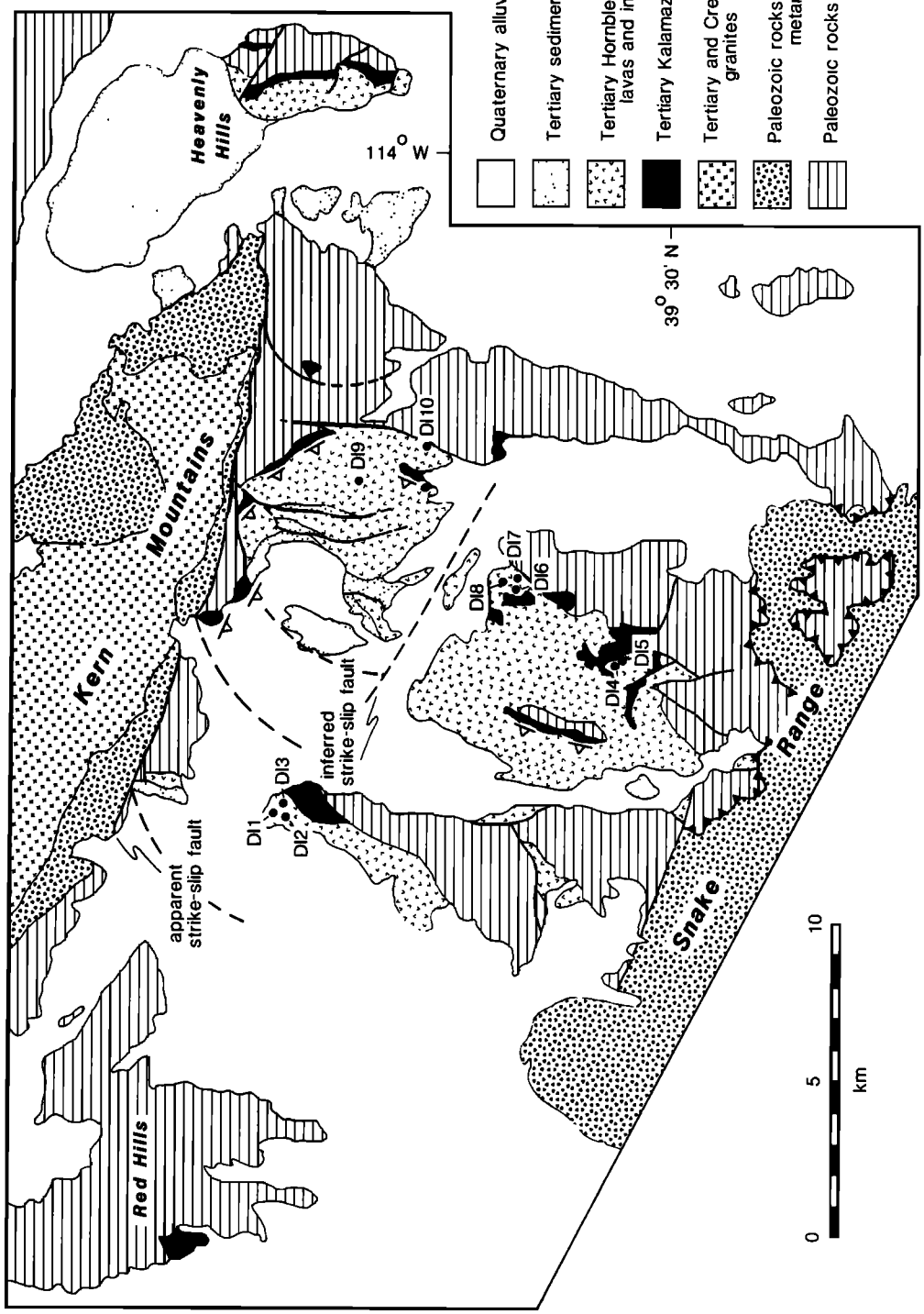


Fig. 2. Geologic map of the region between the Kern Mountains and the northern Snake Range. Dots and labels indicate paleomagnetic sampling sites in the hornblende dacite intrusion. See Structural Implications section for discussion of the apparent and inferred strike-slip faults.

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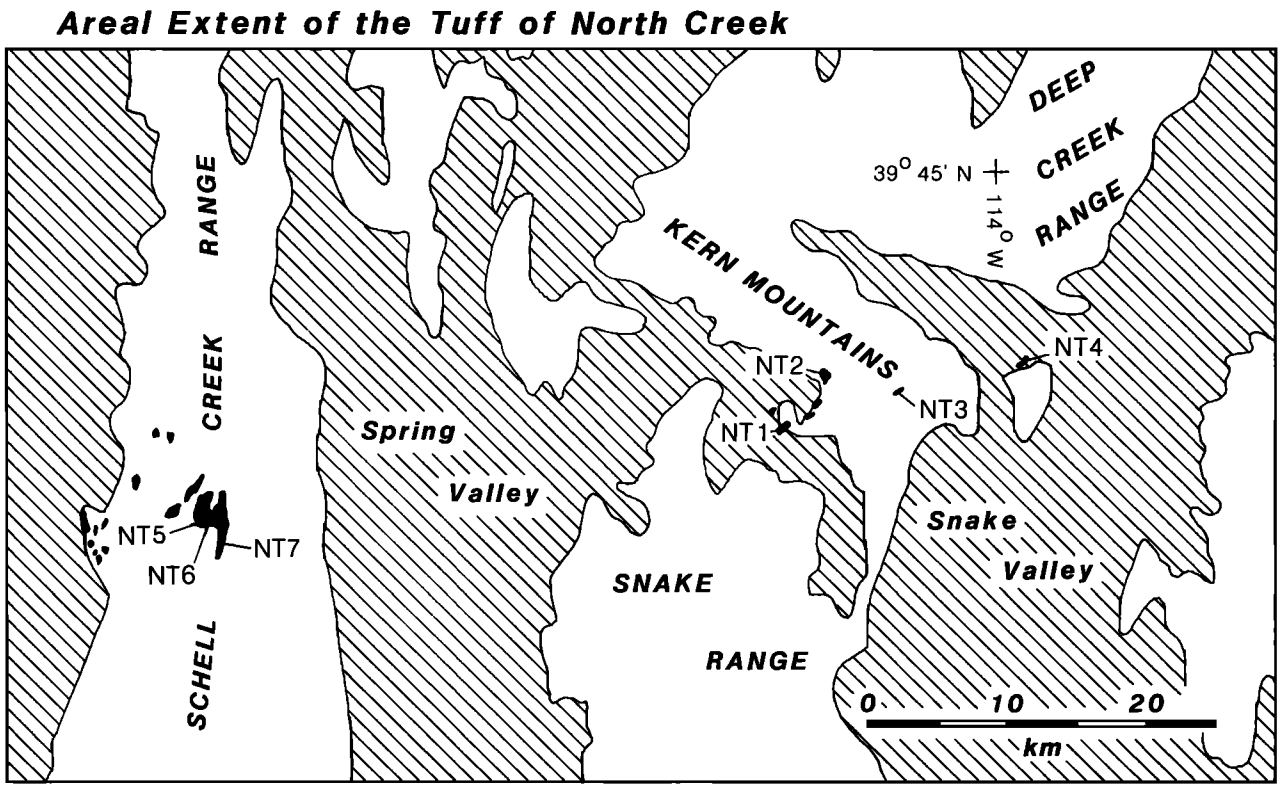


Fig. 3. Regional map of east central Nevada showing exposures of the tuff of North Creek (black areas) and the paleomagnetic sampling sites (labels) within them.

drastically thinned by two generations of east-dipping high-angle normal faults that flattened abruptly into the Snake Range decollement. Each of these episodes of faulting was associated with 40° of westward rotation, so that the first generation faults now dip gently westward, the second generation faults dip gently eastward, and locally bedding is near vertical. Lower plate rocks beneath the Snake Range decollement were thinned and extended an equivalent amount coaxial with the upper plate rocks suggesting that significant movement along the Snake Range decollement need not have occurred [*Gans and Miller*, 1983].

METHODS

Samples for paleomagnetic analysis were collected from sites in the Kalamazoo Tuff, hornblende dacites, and the tuff of North Creek (Tables 1 and 2) in eastern Nevada and western Utah (Figures 1–3) as 2.5-cm drill cores using a portable gasoline-poleward drill. The cores were oriented in the field, with an estimated accuracy of 2°, using magnetic and solar compasses and a clinometer. Several attitude measurements of the eutaxitic structure were also made at each site in the tuffs. Difficulties in making these measurements were often caused by poor outcrop exposure (only an apparent dip on a single outcrop surface could be measured) and by surface weathering and lichen growth which obscured the flattened pumice structures. These attitude measurements sometimes varied widely indicating they were possibly the source of significant error. For this reason attitudes of the eutaxitic structure at each site in the Kalamazoo Tuff and tuff of North Creek were determined in the laboratory from the core samples by measuring the plane of the eutaxitic structure in each of the samples of a given site. The poles to these planes were corrected from sample coordinates to geographic coordinates using the core orientation data. The poles were then averaged and the strike and dip of the eutaxitic structure was calculated from the mean pole (Figure 4). These strikes and dips for sites in the Kalamazoo Tuff and tuff of North Creek are listed in Table 1. Structural corrections were made by a simple rotation of the plane of the eutaxitic structure to the horizontal about the strike axis.

Cylindrical specimens 2.2 cm in length were cut from the core samples and were subjected to either progressive alternating field (af) or thermal demagnetization. Af demagnetization in peak fields between 1 and 100 millitesla (mT) was the primary technique used in this study. Stepwise thermal demagnetizations of 1–3 specimens from 17 sites in the Kalamazoo Tuff were performed in air to temperatures in excess of 600°C employing a shielded furnace. The remanent magnetization of individual specimens was measured with a cryogenic magnetometer. Demagnetization techniques were successful in isolating characteristic components of magnetization for the rock samples from all but one site (DI1). At this site inconsistent lightning-induced remanence is presumed to overprint the characteristic component of magnetization. The characteristic components form linear segments on orthogonal vector plots (Figure 5) which decay toward the origin with increased demagnetization. Least squares lines were fitted to these linear segments employing principal component analysis [*Kirschvink*, 1980]. Results for those samples having two components with overlapping coercivity spectra define great circle arcs on equal area plots with increased demagnetization and required the use of

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demagnetization planes [*Halls*, 1976; *Kirschvink*, 1980] to determine the characteristic magnetization. The statistical methods of *Fisher* [1953] are reported for the single-component site means and the overall paleomagnetic direction for the hornblende dacite intrusion. Bingham statistics [*Onstott*, 1980] were used to calculate the mean directions for the site (DI1) exhibiting converging demagnetization planes (Table 2) and for the overall paleomagnetic directions of the Kalamazoo Tuff and the tuff of North Creek.

SAMPLING AND RESULTS

Kalamazoo Tuff

The Kalamazoo Tuff, primarily unit B, was sampled at 39 sites across a 100-km wide region from the Egan Range in the west to the Confusion Range in the east (Figure 1). North-south transects from 5 to 50 km in length were possible in exposures of the tuff in the Schell Creek Range, in the region south of the Kern Mountains, and in the Heavenly Hills. The latter two

TABLE 1. Site Mean Directions and Poles for the Kalamazoo Tuff and the Tuff of North Creek

	Uncorrected			Corrected						Pole	
Sitet	Inclination, degrees	Declination, degrees	Strike Dip	Inclination, degrees	Declination, degrees	No. of Samples	R	k	α_{95}	Latitude, degrees	East Longitude, degrees
					Kalamazoo Tuff						
KT1	−12.5	17.4	149.2/105.0	49.4	330.9 e	9	8.7740	35	8.8	64.5	143.4
KT2	41.0	27.1	155.8/56.5	58.6	310.6 e	9	8.9740	308	2.9	52.4	171.8
KT3	30.0	21.3	150.7/64.2	55.1	314.6 e	9	8.9626	214	3.5	54.2	164.2
KT4	61.9	18.5	180.6/47.3	44.8	309.8 e	8	7.9816	380	2.9	46.5	154.0
KT5	72.6	272.1	262.0/18.1	63.0	311.6 e	10	9.9754	365	2.5	54.3	180.2
KT6	46.8	83.5	181.4/75.8	56.8	281.3 e	10	9.9621	237	3.1	30.4	181.3
KT7[a]	63.2	342.0	223.6/64.0	2.0	326.0	10	9.9723	325	2.7	40.5	113.3
KT8[a]	64.6	335.0	114.7/25.9	69.0	270.8	10	9.9901	911	1.6	30.8	200.8
KT9	49.5	333.1	121.7/4.8	51.9	327.9 e	9	8.7660	34	8.9	63.3	150.6
KT10	51.7	84.1	198.7/48.9	70.1	338.1	10	9.9514	185	3.6	69.3	207.6
KT10[b]	51.7	84.1	196/64	59.4	313.0 e	10	9.9514	185	3.6	54.4	172.4
KT11	57.2	256.1	267.1/20.6	55.4	287.6 e	11	10.9193	124	4.1	34.3	176.8
KT12[c]	72.7	77.9	155.7/49.5	57.2	239.1	11	10.8890	90	4.8	4.5	202.8
KT13[d]	52.4	329.8	170.6/11.3	47.3	317.7	10	9.9536	194	3.5	53.6	150.8
KT14	70.4	52.7	272.7/0.6	70.0	51.4	9	8.9231	104	5.1	52.9	295.1
KT14[b]	70.4	52.7	225/17	66.1	10.0 w	9	8.9231	104	5.1	78.6	281.0
KT15	70.2	30.5	225.1/17.3	60.8	357.3 w	10	9.9956	2024	1.1	87.0	204.0
KT16[c]	48.9	69.2				10	9.8077	47	7.1	56.7	14.9
KT17[d]	52.8	337.7	194.9/17.6	40.4	324.2	8	7.9929	989	1.8	55.5	137.2
KT18	65.3	45.6	186.8/32.0	65.4	328.3 w	10	9.9468	169	3.7	66.0	184.7
KT19	71.2	338.1	187.7/21.4	55.4	307.2 w	11	10.9844	642	1.8	48.8	167.5
KT20	40.8	310.9	73.6/44.7	65.8	252.5	10	9.9871	697	1.8	18.7	203.1
KT20[b]	40.8	310.9	350/25	52.6	335.1 w	10	9.9871	697	1.8	69.0	145.2
KT21[e]	75.5	135.3	263.3/47.4	53.2	8.2	9	8.7856	37	8.5	81.2	14.5
KT22	47.8	308.2	43.1/10.9	58.6	306.8	10	9.9553	201	3.4	49.6	172.6
KT22[b]	47.8	308.2	350/25	59.4	339.5 w	10	9.9553	201	3.4	74.3	164.2
KT23	45.8	307.7	358.7/18.5	58.4	325.6	10	9.9653	260	3.0	63.6	165.4
KT23[b]	45.8	307.7	350/25	58.0	336.7 w	10	9.9653	260	3.0	72.0	159.9
KT24	50.7	307.1	339.7/19.0	57.4	332.2	10	9.9918	1096	1.5	68.4	160.2
KT24[b]	50.7	307.1	350/25	62.1	342.2 w	10	9.9918	1096	1.5	76.2	177.1
KT25	47.3	81.7	206.2/56.6	60.6	347.7 w	10	9.8683	68	5.9	80.4	171.9
KT26	59.1	63.2	165.6/47.3	71.9	276.4	10	9.9915	1061	1.5	35.4	203.5
KT26[b]	59.1	63.2	195/55	53.7	320.3 w	10	9.9915	1061	1.5	58.1	158.6
KT27	72.7	63.3	189.0/36.8	65.4	303.6 w	9	8.9129	92	5.4	49.4	185.4
KT28	77.7	285.9	245.8/29.1	51.9	320.5 e	10	9.9861	649	1.9	57.6	156.3
KT29[a]	50.9	299.3	35.6/36.2	85.2	250.6	3	2.9827	115	11.5	35.8	234.7
KT30[a]	64.7	296.0				6	5.9652	144	5.6	44.0	186.6
KT31	53.9	51.1	23.7/43.4	23.6	78.9	8	7.9574	164	4.3	16.3	333.4
KT31[b]	53.9	51.1	180/55	57.0	312.9 e	8	7.9574	164	4.3	53.6	168.3
KT32	68.6	100.6	184.1/38.7	72.4	266.3	8	7.9908	758	2.0	30.8	207.5
KT32[b]	68.6	100.6	201/55	55.9	297.8 e	8	7.9908	758	2.0	41.9	173.1
KT33[c]	60.7	72.8				9	8.7405	31	9.4	36.5	308.0
KT34[d]	72.3	353.0	207.0/36.6	41.6	316.7	6	5.9869	381	3.4	50.4	145.0
KT35[d]	72.1	1.8	174.0/43.8	45.7	289.8	6	5.9961	1291	1.9	31.6	166.2
KT35[b]	72.1	1.8	180/48	40.1	293.7	6	5.9961	1291	1.9	32.2	159.7
KT36[d]	69.5	5.4	202.8/61.6	20.7	313.7	8	7.9786	327	3.1	39.9	133.2
KT36[b]	69.5	5.4	180/48	40.6	297.4	8	7.9786	327	3.1	35.2	157.8
KT37[d]	59.8	33.8	214.6/53.6	30.5	340.3	7	6.9882	507	2.7	61.1	107.6
KT38	59.5	61.8	216.6/44.1	50.3	352.5 w	7	6.9808	312	3.4	79.5	103.4
KT39	59.4	51.4	223.0/43.7	42.3	356.0 w	8	7.9927	956	1.8	74.6	79.1
Avg 1				62.8	320.6	25	[−13/−7]	[6.0/8.4]		60.6	177.5
Avg 2				58.9	324.4	25	[−30/−13]	[3.7/5.9]		62.8	167.1
Avg east				55.8	310.5	12	11.8417	70	5.2		
Avg west				59.2	338.7	13	12.7354	45	6.2		

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TABLE 1. (continued)

Site	Uncorrected Inclination, degrees	Uncorrected Declination, degrees	Strike Dip	Corrected Inclination, degrees	Corrected Declination, degrees	No. of Samples	R	k	α_{95}	Pole Latitude, degrees	Pole East Longitude, degrees
				North Creek Tuff							
NT1	77.1	66.0	246.1/37.6	50.5	356.7	8	7.9646	198	4.0	81.3	84.9
NT2	−4.7	19.9	145.3/54.5	37.8	8.4	9	8.9806	412	2.5	70.2	42.2
NT3	9.0	38.0	182.3/50.8	33.1	19.1	8	7.9796	343	3.0	62.8	22.9
NT4	−18.9	49.3	184.3/106.5	47.2	14.5	6	5.9607	127	7.2	22.8	11.9
NT5	41.1	352.9	125.4/21.9	54.6	333.4	7	6.9876	486	2.7	68.5	152.3
NT6	46.5	3.9	154.6/22.1	53.0	338.4	6	5.9860	358	3.6	71.7	143.1
NT7	56.0	345.3	159.8/25.0	50.7	311.4	8	7.9944	1254	1.6	50.1	159.5
Avg				49.1	354.5	7	[−161/−6]	[3.1/16.7]		79.4	92.6

R, vector sum; k, precision parameter [*Fisher*, 1953]; α_{95}, radius of 95% confidence in degrees; Avg, mean paleomagnetic directions calculated (1) using core-measured attitudes only and (2) using best correction to estimate direction of the geomagnetic field (see Discussion); e, east sites near the Kern Mountains and east; w, west sites in the Schell Creek Range and west. For the mean paleomagnetic directions with Bingham statistics [*Onstott*, 1980], R is not calculated, $[k_1/k_2]$ are the two Bingham concentration parameters, and $[\alpha_1/\alpha_2]$ are the two Bingham 95% confidence limits about the mean direction.

[a]Remagnetized by hornblende dacite intrusion.
[b]Strike/dip is that of nearby sediments or regional attitude of tuff units.
[c]Eutaxitic structure indicates rheomorphic flow.
[d]Vitrophyre.
[e]Site is in a slump block.

transects were made to determine whether large map scale deflections in the strikes of the tilted sections was a consequence of oroclinal bending due to strike-slip faulting (i.e., vertical axis rotations) or whether they were produced by some other mechanism. Two vertical sections of the tuff, approximately 180 and 88 m thick, were sampled in the type area at Kalamazoo Creek (KT13-15 and KT37-39, respectively). In each of these sections, sites were located in the basal vitrophyre of unit B (KT13, KT37), in the devitrified portion of unit B (KT14, KT38), and in unit C (KT15, KT39) to determine if secular variation of the geomagnetic field occurred between cooling and magnetization of these two units. Two sites (KT35-36) were also sampled at Kalamazoo Creek in vitrophyre where the entire tuff thinned to about 10 m over a topographic high in tilted Mississippian limestone basement. The eutaxitic structure of the tuff conforms with the slope of this basement high and had an initial dip of approximately 20° on its flanks (Table 1). At Schellbourne Pass in the northern Schell Creek Range samples were collected from sites in strongly folded devitrified unit B (KT16, KT33) and from the apparently undeformed unit B basal vitrophyre (KT34). Sampling at site KT33 was distributed around a single fold to test whether or not magnetization acquisition predates deformation within the unit. Three samples were taken from unit B (KT29) within a meter or two of its contact with a hornblende dacite intrusion (DI5) near the northern end of the Snake Range (Figure 1) to determine if the tuff had been remagnetized by the younger intrusion.

Intensities of natural remanent magnetization (NRM) for samples of the Kalamazoo Tuff are generally around 1 A m^{-1}. At several sites, samples with significantly higher natural remanent magnetization (NRM) intensities tend to show more dispersed NRM directions suggesting that the NRM at these

TABLE 2. Site Mean Directions and Poles for the Hornblende Dacite Intrusion

Site	Uncorrected Inclination, degrees	Uncorrected Declination, degrees	No. of Samples	R	k (k_1/k_2)	α_{95} (α_1/α_2)	Pole Latitude, degrees	Pole East Longitude, degrees
DI1	73.9	293.8	5		−53/−2	7.3/12.0	45.0	205.4
DI2	58.3	284.8	7	6.9397	100	6.1	33.6	181.2
DI3	43.7	288.9	6	5.9862	363	3.5	30.0	165.4
DI4	61.4	319.5	8	7.9521	146	4.6	59.6	174.8
DI5	53.4	286.3	7	6.9767	257	3.8	32.3	175.4
DI6	73.6	278.6	6	5.9868	378	3.5	37.4	206.7
DI7	56.9	310.7	8	7.9524	147	4.6	51.9	168.8
DI8	73.8	301.1	5	4.9116	45	11.5	48.7	205.3
DI9	75.8	85.7	7	6.9893	563	2.6	36.5	279.9
DI9*	60.6	287.1	7	6.9893	563	2.6	36.3	183.3
DI10	63.6	81.2	8	7.9929	989	1.8	32.3	301.5
DI10*	70.4	305.3	8	7.9929	989	1.8	51.0	197.1
Avg	63.2	295.3	10	9.8103	47	7.1	43.0	184.3

Abbreviations are as in Table 1. For directions resolved using demagnetization planes and Bingham statistics [*Onstott*, 1980], R is not calculated, k_1/k_2 are the two Bingham concentration parameters for poles to the demagnetization planes, and α_1/α_2 are the two Bingham 95% confidence limits about the mean direction.

*Corrected for a strike and a dip of 190°/43° (see text); included in average calculation instead of uncorrected direction.

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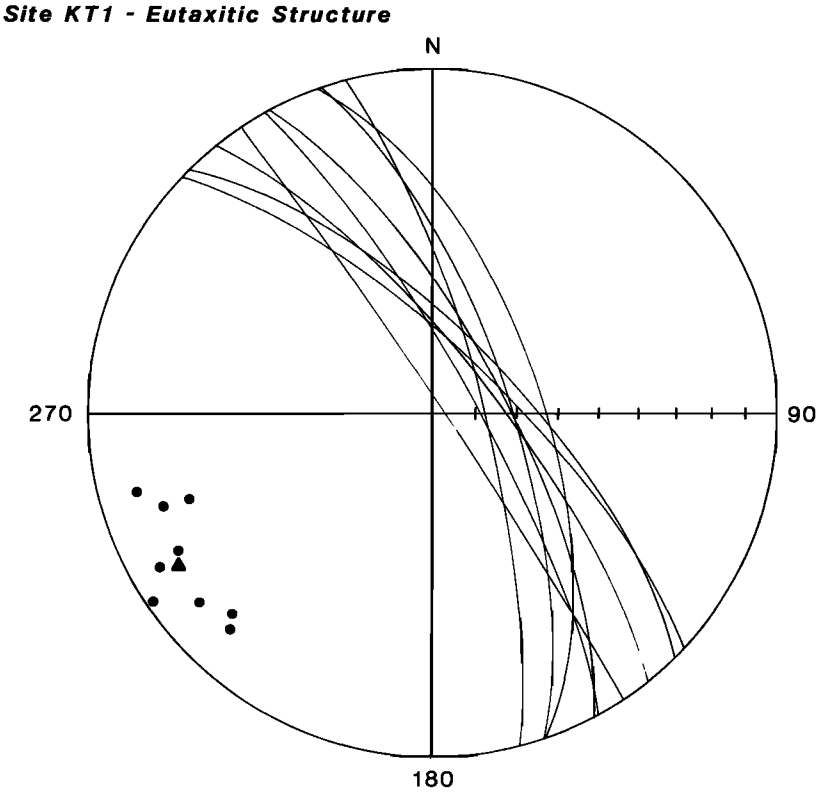


Fig. 4. Equal area plot of planes (curved lines) and their poles (dots) measured from the eutaxitic structure within individual core samples from site KT1 in the Kalamazoo Tuff [of *Gans and Miller*, 1983]. The triangle indicates an average pole used to calculate the structural correction for the site.

sites has been affected by lightning strikes. Progressive af demagnetization was successful in removing the lightning-induced magnetizations and defining a consistent higher-coercivity (generally > 20 mT) component of magnetization (Figure 5*a*). Thermal demagnetization experiments on samples from the Kalmazoo Tuff (Figure 5*b*) yield characteristic components of magnetization, with maximum laboratory unblocking temperatures of about 630° C, identical to those determined with af treatments.

Site mean directions for the Kalamazoo Tuff before and after correction for tilt of the eutaxitic structure are shown in Figures 6 and 7 and are listed in Table 1. The directional dispersion is generally reduced by applying tilt corrections, but the dispersion remains larger than expected, and for a number of sites the correction is clearly erroneous. As discussed below, the large scatter in the corrected data for the Kalamazoo Tuff is caused by initial deviations from the paleohorizontal as defined by the eutaxitic structure due to irregular topography and rheomorphic flow. Other possible sources of error include the occurrence of geomagnetic secular variation during emplacement and cooling of the tuff units and remagnetization of the tuff at sites close to a younger hornblende dacite intrusion.

Hornblende Dacite Intrusion

The large hornblende dacite intrusion exposed in the low region between the Kern Mountains and the northern Snake Range was sampled at 10 sites (Figure 2). These sites were located near roof pendants of Kalamazoo Tuff suspected of having been reheated and remagnetized by the intrusion. Other localities were distributed across its area of exposure to document any postemplacement deformation of the unit. Samples were collected from several sites at each locality to better estimate the relative effects of secular variation versus structural disturbance on the site mean directions.

NRM intensities for samples from the dacite intrusion range from about 1 to 10^{-1} A m^{-1}. All samples from site DI1, and a number of samples from three other sites, have high NRM intensities (1 A m^{-1}) and dispersed NRM directions that may be indicative of lightning-induced magnetizations. Generally, af demagnetization was sufficient treatment to isolate the characteristic magnetizations (Figure 5*c*), but demagnetization planes analysis was also required for samples from site DI1 to determine its mean direction.

The in situ site mean directions for the hornblende dacite intrusion are shown in Figure 8 and listed in Table 2. With the exception of sites DI9 and DI10, the directions are consistent across the intrusion and group near the corrected directions for sites in the Kalamazoo Tuff (Figures 6*b* and 7*b*). A structural correction (strike = 190°, dip = 43°) for sites DI9 and DI10, estimated from nearby units overlying the dacite intrusion, brings then into agreement with the other uncorrected directions for the intrusion (Figure 8).

Tuff of North Creek

The tuff of North Creek was sampled at seven sites within its two principal areas of exposure: along the southern and eastern margins of the Kern Mountains (NT1-4) and near Kalamazoo Summit in the Schell Creek Range (NT5-7, Figure 3). Because of the uncertainties in correlating individual cooling units, more than one cooling unit was probably sampled.

Demagnetization of the tuff of North Creek was for the most part straightforward. NRM intensities were generally around 1

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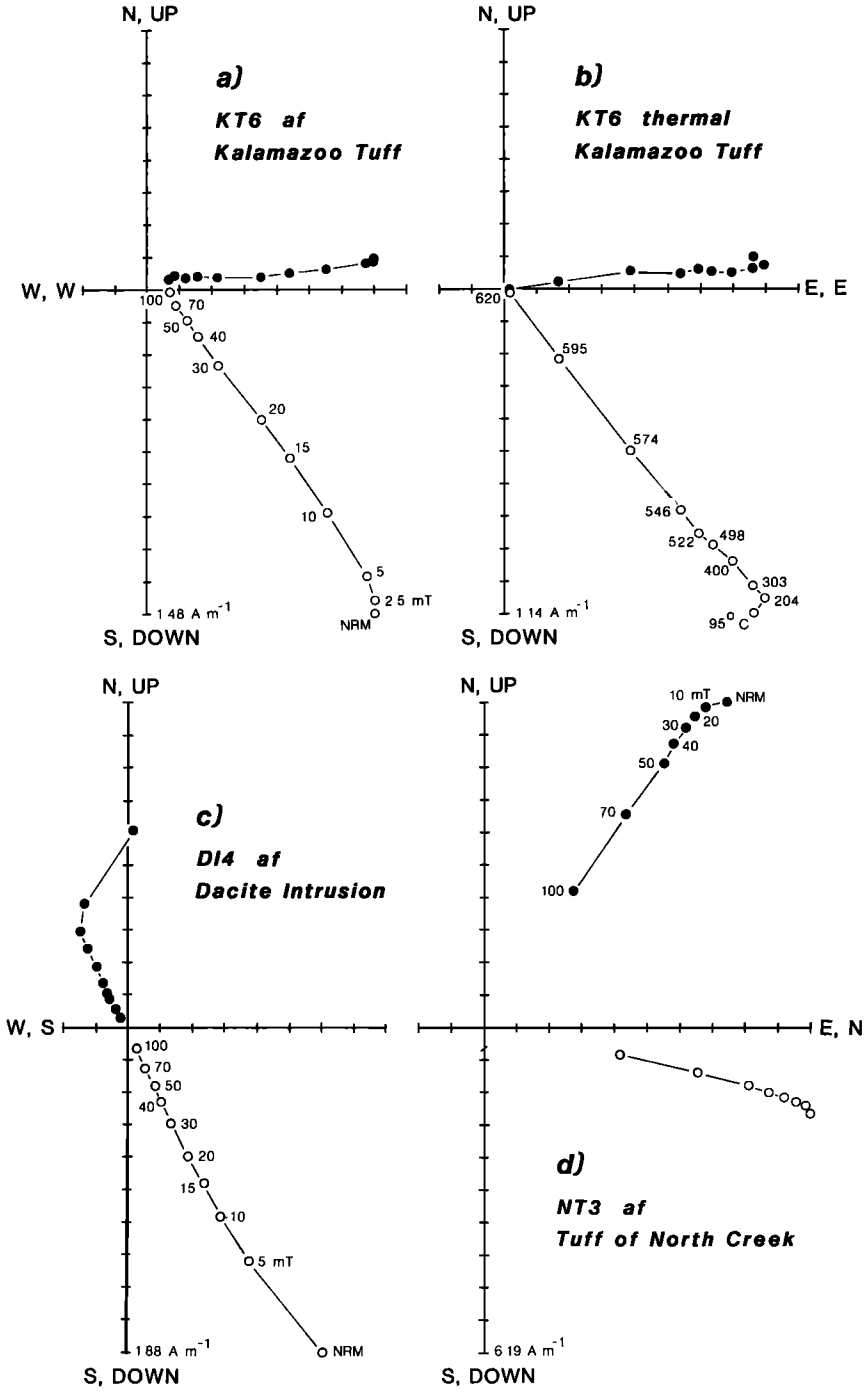


Fig. 5. In situ orthogonal projections of demagnetization vector endpoints. Figures 5*a* and 5*b* show similar results obtained with alternating field and thermal demagnetization, respectively, of samples from site KT6; Figure 5*c* shows af demagnetization of a sample from site DI4 in the hornblende dacite intrusion; and Figure 5*d* shows af demagnetization of a sample from site NT3 in the tuff of North Creek. Solid symbols indicate projections of vector endpoints onto the horizontal plane and open symbols onto the vertical plane.

A m^{-1}. Sites NT1 and NT2 had higher NRM intensities (10–100 A m^{-1}) and scattered NRM directions suggestive of lightning-induced magnetizations. Af demagnetization (>30 mT) was successful in removing the lightning-induced magnetizations and in isolating a stable characteristic magnetization for all samples (Figure 5*d*). In situ and corrected site mean directions are given in Table 1 and shown in Figure 9. The variation in inclinations is reduced after correction for structural tilt and the remaining dispersion in declination (Figure 9*b*) probably reflects between-site vertical axis rotations. However, secular variation between emplacement of the individual cooling units could also be a cause of the dispersion.

Discussion

Sources of Dispersion

The internal consistency of both sample and site mean directions (Tables 1 and 2) indicates that the Kalamazoo Tuff and related rock units are most likely accurate recorders of the ancient geomagnetic field. The basic improvement in

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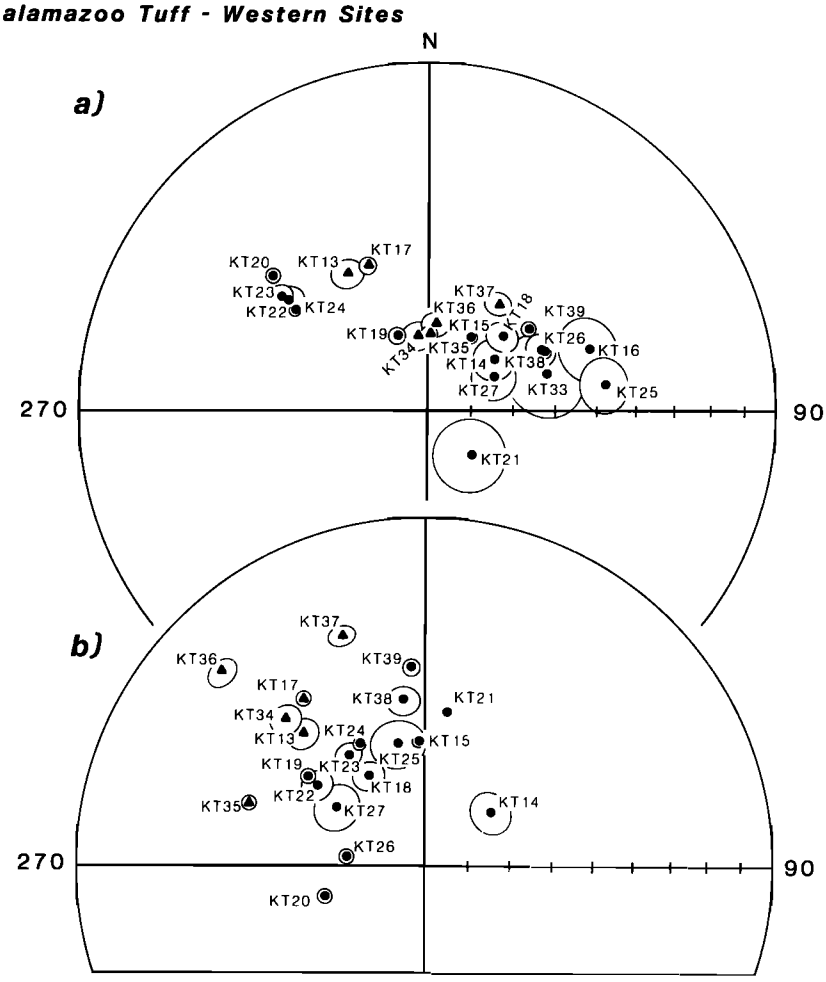


Fig. 6. Mean directions for sites in the Kalamazoo Tuff [of *Gans and Miller*, 1983] from the Schell Creek Range and to the west, (*a*) uncorrected and (*b*) corrected for structural tilt based on eutaxitic structures, with 95% confidence limits (Table 1). Triangles indicate sites in the unit B vitrophyre. The corrected declination for site KT21 is probably incorrect because the site is in a slump block. Diagrams are equal area projections where solid symbols indicate projection from the lower hemisphere.

dispersion of the site mean directions for the Kalamazoo Tuff after structural correction ($k_{\text{in situ}} = 6$, $k_{\text{corrected}} = 33$; Figures 6 and 7) demonstrates that at most locations its eutaxitic structure was approximately horizontal at the time the tuff was magnetized. The tilt-corrected variation in inclination, however, is much greater than expected. If the tuff (unit B) had been uniformly magnetized at a point in time when the eutaxitic structure was consistently horizontal, then the corrected directions should have constant inclinations and the between-site differences in declination could represent between-site vertical axis rotations.

Part of the variation in inclination of corrected site mean directions for the Kalamazoo Tuff is probably due to secular variation of the geomagnetic field during the tuff's emplacement and cooling. The mean directions for sites in the unit B vitrophyre, which most likely represent the earliest remanence acquired, are consistently shallower in inclination by about 20° than those in the devitrified portions of units B and C (Figure 6*b*, Table 1). Other explanations for this difference in inclination include postblocking rheomorphic flow of the tuff and anisotropy control on the remanent direction related to extreme compaction. However, the vitrophyre for the most part is not densely welded, and the same direction is observed in the tuff where it thins to a single vitrophyre (≈10 m thick) over a topographic high in the Paleozoic basement rocks. In addition, the flattened pumice fragments within the vitrophyre show no evidence of lineation. Similarly, small but significant differences in directions are observed between the devitrified portions of units B and C at sites KT14-15 and KT38-39, which may indicate that these units also cooled through their magnetization blocking temperatures at different times in slightly different ambient fields.

Initial variations in attitude of the eutaxitic structures also contribute to the dispersion of the tuff's corrected site mean directions. This is especially well-illustrated by sites KT35 and KT36 where the eutaxitic structure conforms to a hillslope in the paleotopography. This difference in attitude causes the mean directions for these two sites to diverge widely after the structural correction is applied (Figure 6). Similarly, sites located along the western edge of the Schell Creek Range (KT20, 22, 23, and 24; Figure 1) have approximately the same in situ directions ($k = 350$) and dispersed corrected directions ($k = 24$) that imply an initial variation in the eutaxitic structure (Figure 6). Two other sites (KT14, 31) show divergent corrected directions (Figures 6 and 7) near sites that do not, suggesting that the eutaxitic structures may vary significantly over small distances due to differential compaction or rheomorphic flow of the tuff.

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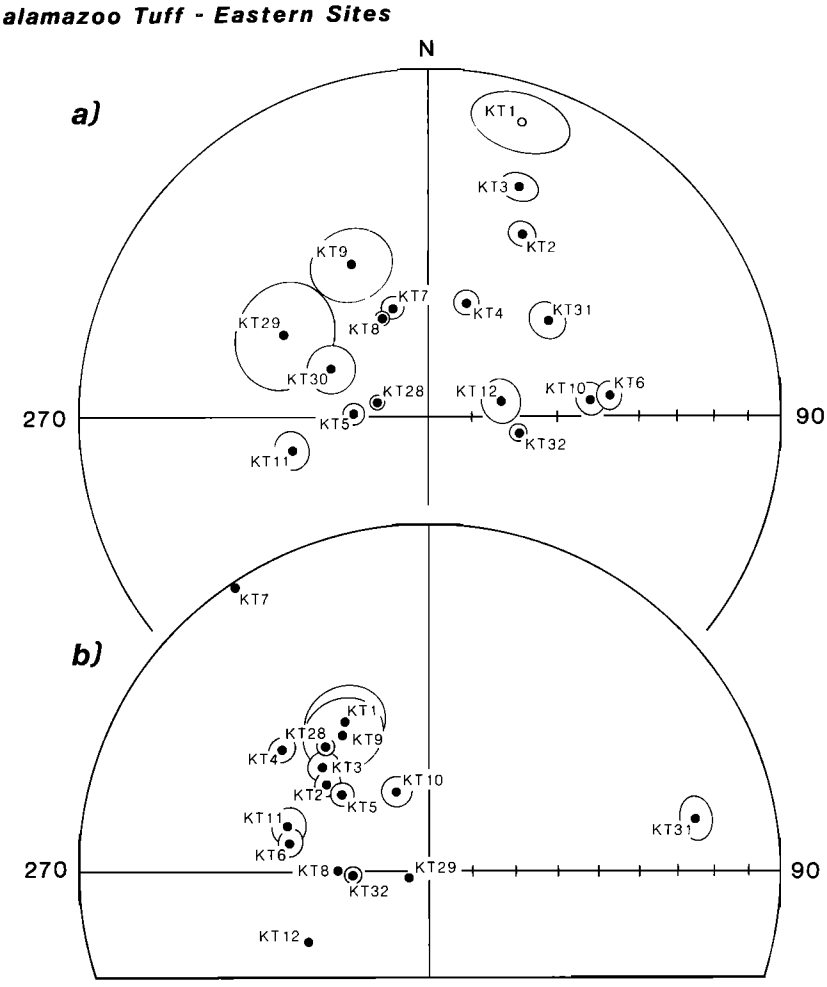


Fig. 7. Site mean directions for the Kalamazoo Tuff [of *Gans and Miller*, 1983] from south of the Kern Mountains and to the east, (*a*) uncorrected and (*b*) corrected for structural tilt based on eutaxitic structures, with 95% confidence limits (Table 1). Corrected means without confidence limits are incorrect because of remagnetization at these sites by the younger hornblende dacite intrusion (see Discussion). Diagrams are equal area projections where solid symbols indicate projection from the lower hemisphere and open symbols from the upper hemisphere.

Rheomorphic flow was indicated at three sites in the Kalamazoo Tuff (KT12, 16, 33) by a stretching out of the fiamme which gives the tuff a banded and lineated appearance. Aspect ratios of flattened pumice fragments greater than 25 imply deformation beyond the simple loss of porosity due to compaction [*Ragan and Sheridan*, 1972]. At site KT33 the banded fiamme have been folded and the in situ paleomagnetic data group according to position in the fold. It is clear from the similarity of directions that the tuff was magnetized after folding had occurred; corrections measured from the banded fiamme, when applied, disperse the magnetization directions (Figure 10). However, the distinct grouping of in situ directions for samples from the two limbs and axial region of the fold indicates that deformation continued after acquisition of the tuff's remanent magnetization. Assuming rigid body deformation of a cylindrical fold, this would amount to about 20% unfolding as the tuff continued to cool.

The corrected site mean directions for the Kalamazoo Tuff at three sites (KT7, KT29, KT30; Figures 6 and 7) in the vicinity of the Kern Mountains (Figure 2) are inferred to have been dispersed due to reheating and remagnetization of the tuff by the adjacent hornblende dacite intrusion. Site KT7, situated within a few meters of the contact between the tuff and the intrusion, yields a mean which, upon correction, moves from a direction similar to that of the dacite intrusion to an anomalously shallow northwesterly direction (Figure 7). A nearby site (KT8) has a similar in situ direction and also an improper structurally corrected direction. Sites KT29 and KT30 are from roof pendants of Kalamazoo Tuff in the southern part of the hornblende dacite intrusion. In geographic coordinates, they have directions similar to those for the adjacent exposures of the intrusion, but at site KT30 the eutaxitic structure is not discernible due to recrystallization.

Structural Implications

To achieve the most accurate estimate of the structural deformation affecting the remanence direction of the Kalamazoo Tuff, corrected site means likely to have been influenced by the sources of dispersion discussed above have been either omitted from the data set or corrected using attitudes based on regional trends of the tuff or nearby stratified units, rather than the locally inconsistent eutaxitic structures (Table 1). Figure 11*a* shows corrected site means for the tuff excluding those from the unit B vitrophyre, those thought to be remagnetized by the dacite intrusion, and

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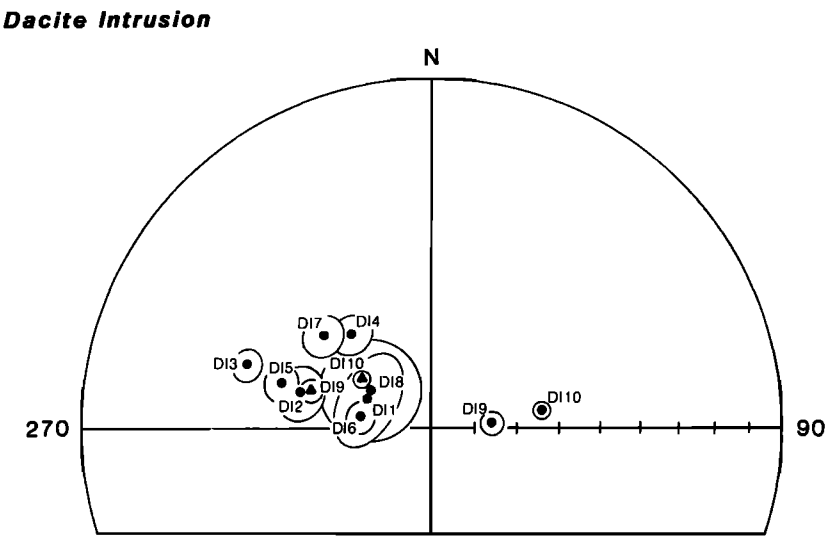


Fig. 8. Equal area plot of site mean directions with 95% confidence limits for the hornblende dacite intrusion. Circles indicate in situ directions, and triangles indicate structurally corrected directions using the attitude of sedimentary units.

those for sites exhibiting rheomorphic flow features. Site KT21 also has been omitted because its mean direction most likely reflects movement associated with recent slumping rather than middle Tertiary extension (Figure 6).

The corrected mean directions for the Kalamazoo Tuff from sites in the Schell Creek Range and to the west (Figure 1) tend to have more easterly declinations (Figure 11*a*) than those for sites near the Kern Mountains and to the east. These differences in declination between geographic locations could be attributed to the ancient nondipole field. However, comparisons of field directions over broad geographic areas based on a modern reference field [*Shuey et al.*, 1970] suggest that the distances between sites in this study are too small for local directions to be significantly influenced by nondipole effects. Mean directions for the Kalamazoo Tuff in these two regions (Figure 11*b*) could indicate a clockwise rotation of the Schell Creek Range 28° ± 12° [*Beck*, 1980; *Demarest*, 1983] relative to the Kern

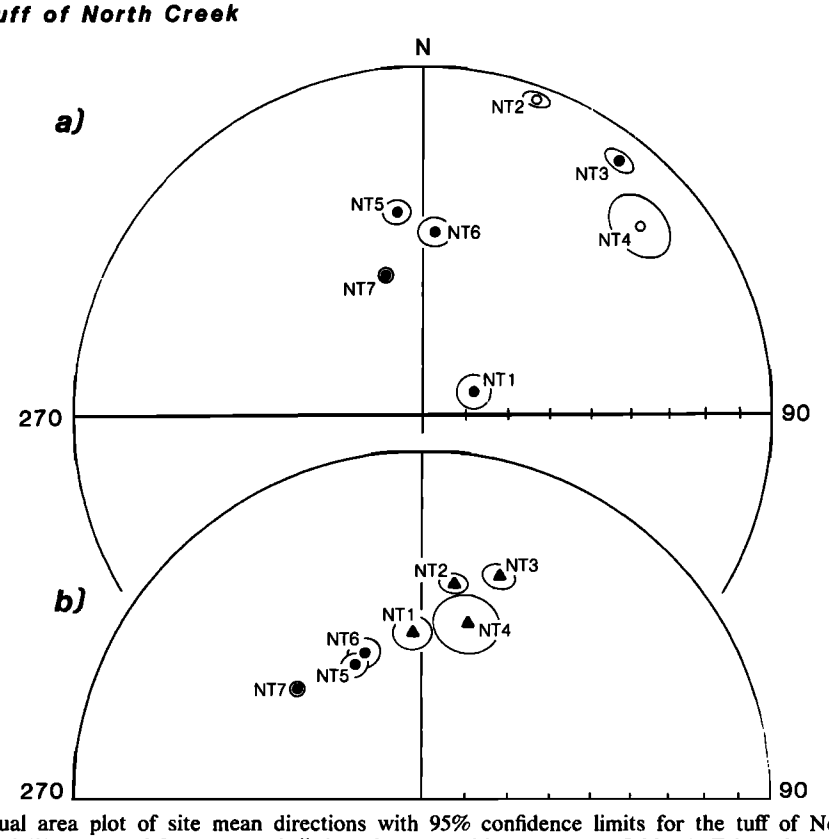


Fig. 9. Equal area plot of site mean directions with 95% confidence limits for the tuff of North Creek, (*a*) uncorrected and (*b*) corrected for structural tilt based on eutaxitic structures (Table 1). Triangular symbols indicate eastern sites and circles indicate western sites (see Figure 3).

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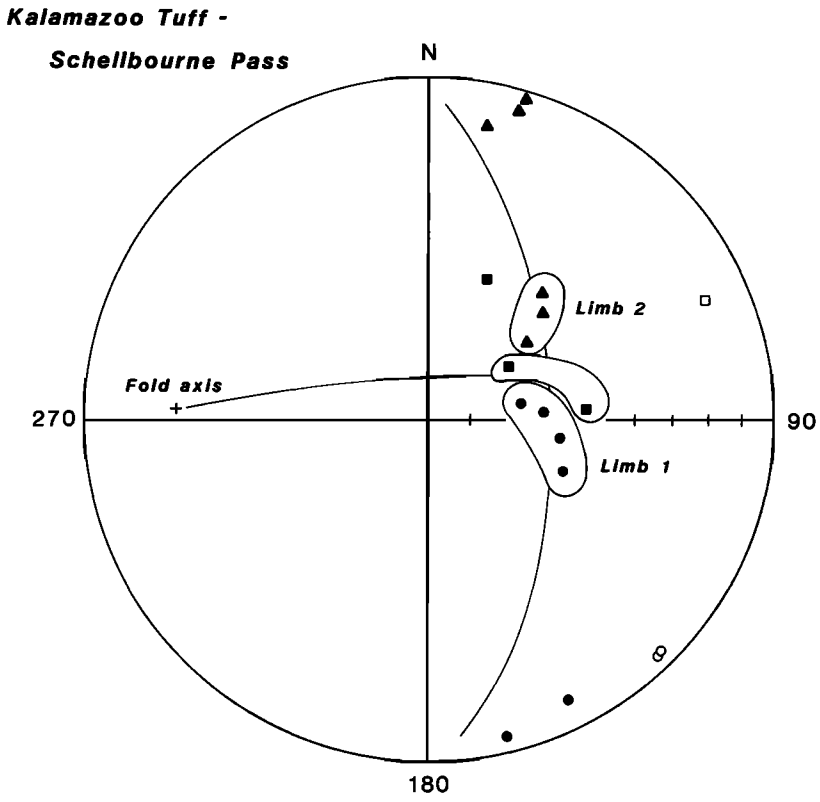


Fig. 10. Equal area plot of directions for samples from around a fold in the Kalamazoo Tuff [of *Gans and Miller*, 1983] at site KT33 near Schellbourne Pass. Deformation of the eutaxitic structure at this locality indicates that the tuff has undergone rheomorphic flow. Enclosed directions (limb 1, limb 2) are uncorrected for folding, and the unenclosed directions (coded by symbol) are corrected for folding. The fold axis is determined assuming a cylindrical fold and the plane perpendicular to the fold axis is also drawn. Solid symbols indicate projections from the lower hemisphere and open symbols from the upper hemisphere.

Mountains region, although averaged site mean directions for the overlying tuff of North Creek (Figure 9, Table 1) suggest the opposite: rotation of the Schell Creek Range 43° ± 22° counterclockwise relative to the Kern Mountains region (Figure 3). These rotations, if reliable, would imply that the Schell Creek Range was rotated about 70° clockwise (relative to the Kern Mountains region) before, and 40° counterclockwise after, emplacement of the tuff of North Creek. Because of nearly identical K-Ar ages for the Kalamazoo Tuff and the tuff of North Creek, such an interpretation seems unlikely. However, it is difficult to dismiss the data for the Kalamazoo Tuff; the 30° differential rotation is indicated by results from 25 sites distributed over a broad area within a single cooling unit. In contrast, the 40° counterclockwise rotation implied by the results for the tuff of North Creek is based on only seven sites that probably represent separate cooling units. Thus the dispersion of directions and apparent counterclockwise rotation could be

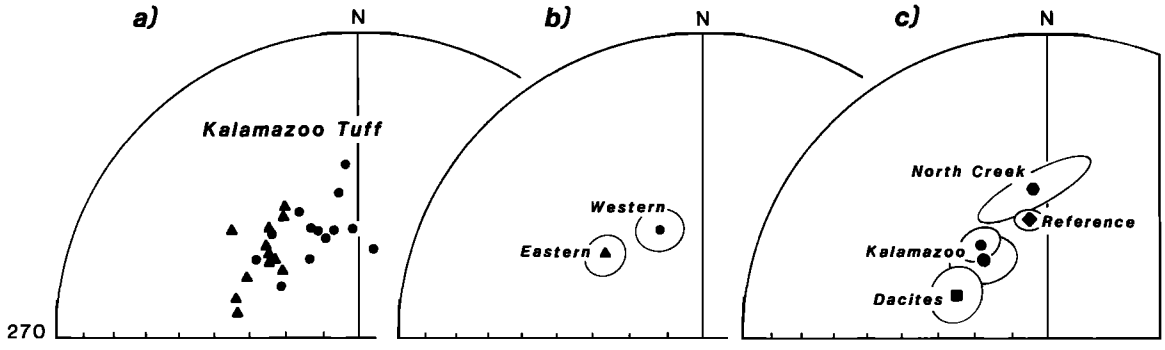


Fig. 11. (*a*) Corrected site mean directions for the Kalamazoo Tuff [of *Gans and Miller*, 1983]. Circles indicate sites from the Schell Creek Range and to the west, and triangles indicate sites from south of the Kern Mountains and to the east. (*b*) Regional averages of the site mean directions shown in Figure 11*a*. (*c*) Unit mean directions for the Kalamazoo Tuff, the hornblende dacite intrusion, and the tuff of North Creek. The larger symbol (with larger 95% confidence limit) for the Kalamazoo Tuff indicates an average using sites corrected for tilt using the measured eutaxitic structures. The smaller symbol for the Kalamazoo Tuff indicates a similar average but with some sites corrected for tilt using the regional attitude of the tuff or that of nearby sediments (Table 1). The reference direction for east central Nevada in Oligocene to early Miocene time is calculated from a pole position given by *Diehl et al.* [1983].

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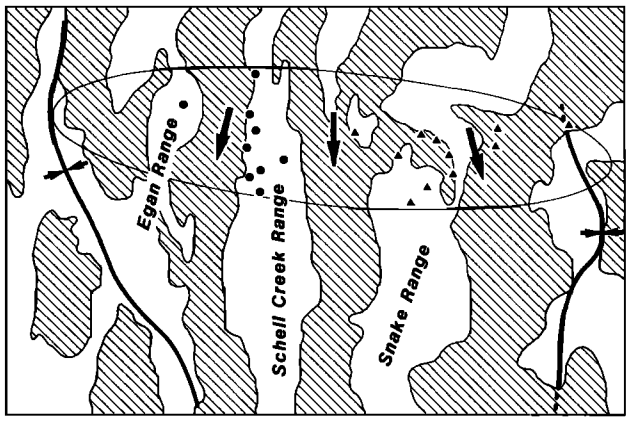


Fig. 12. Map of east central Nevada (including area of Figure 1) showing the distribution of the Kalamazoo Tuff of *Gans and Miller* [1983] (elliptical area) and the distribution of eastern (triangles) and western (circles) sampling sites within the tuff. The three large southward pointing arrows illustrate the apparent 30° rotation between the eastern and western regions. The heavy lines indicate the axes of Mesozoic folds in Paleozoic rocks (Butte synclinorium on the west and the Confusion synclinorium on the east) that are presumed to have been parallel prior to middle Tertiary extension.

caused by geomagnetic secular variation between emplacement of the cooling units of the tuff of North Creek.

The apparent 30° rotation between the eastern and western sampling regions of the Kalamazoo Tuff therefore might suggest differential extension accommodated by distributed strike-slip faulting or N-S shortening. Figure 12 depicts the apparent rotation between sites in the Kalamazoo Tuff within the eastern and western limits of extension as defined by the Confusion and Butte synclinoriums, respectively. These limits suggest, from north to south, a progressive increase in the magnitude of extension to just south of the study area followed by a progressive decrease in the magnitude of extension farther to the south. The increase in extensional strain is also indicated by a higher degree of tilting, a greater density of faulting, and the exposure of deeper structural levels in the area of this study. Tilting in the vicinity is characterized by westward dips generally around 50°–60°; to the north dips are closer to 20° in the Antelope and Cherry Creek Ranges. Ranges farther to the south also have smaller dips, but display the opposite polarity of faulting and tilting. This less extended structural domain to the south is separated from the highly extended domain to the north by a zone of complex block faulting and tilting rather than a discrete transform fault.

The paleomagnetic sampling was also designed to address specific structural problems. One such problem is the apparent oroclinal bending of the Kalamazoo Tuff south of the Kern Mountains (Figure 2). The area between the Kern Mountains and the northern Snake Range is characterized by mutiple generations of variably oriented faults and a large variation in the attitudes of Paleozoic and Tertiary strata. A major E-W trending fault on the southern flank of the Kern Mountains (Figure 2) separates this domain from the structurally more coherent Kern Mountains and Deep Creek Range to the north and has been interpreted by *Gans et al.* [1989] to represent a tear or strike-slip fault which developed in response to differential extention. The large map scale bend in the strike of Paleozoic and Tertiary units near this fault was also interpreted as oroclinal bending due to left lateral movement on the fault. The paleomagnetic data, however, demonstrate that map scale bends in the strike of units (sites KT1-4, Figures 1 and 2) are not a consequence of vertical axis rotations during oroclinal bending. The tilt corrections alone are sufficient to bring the corrected directions into general agreement (Figure 7), regardless of their present strike. The map scale bending of units appears to be the result of southward tilting (normal drag) due to down-to-the-south displacement on the E-W trending fault offsetting sections that had previously been tilted westward (Figure 13). Southward tilting would also explain why the earlier N-S trending, east-dipping normal faults bend in the opposite sense to the sections which they imbricate (Figure 13). This interpretation is also supported by fault-slip indicators on othe E-W trending fault which suggest recent down-to-the-south displacement. The fault probably initially developed as a strike-slip fault in response to differential extension, but was subsequently reactivated as a normal fault. The regional significance of this and perhaps other E-W trending normal faults is unclear. Similar map relations in other areas (e.g., the southern flank of the northern Snake Range [*Grier*, 1984] and the Bullard Peak area of the Harcuvar Mountains [*Reynolds and Spencer*, 1985]) also may reflect superimposed orthogonal sets of normal faults rather than oroclinal bending due to strike-slip faulting.

The age of faulting to the south of the Kern Mountains is also uncertain from map relations. In several localities, the hornblende dacite intrusion appears to immediately postdate much of the faulting: its intrusive contacts apparently truncate many of the faults and tilted sections of Kalamazoo Tuff occur as roof pendants within the intrusion. The paleomagnetic data for the dacite intrusion also support this interpretation. The in situ site mean directions for the intrusion are, for the most part, near the corrected directions for the nearly contemporaneous Kalamazoo Tuff (Figure 8), indicating that the intrusion at these localities is not tilted. However, near sites DI9 and DI10 (Figure 2), the dacite intrusion is imbricated by east-dipping normal faults and overlain by volcanic and sedimentary sequences that dip up to 40° westward. In this area, tilting and faulting are inferred to have continued after emplacement and cooling of the intrusion, because a structural correction based on the attitude of the overlying units is needed for sites DI9 and DI10 to bring them into agreement with the other uncorrected directions for the intrusion (Figure 8). This tilted portion of the hornblende dacite intrusion is probably separated from the undeformed portion of the intrusion to the southwest by a strike-slip fault (Figure 2); early strike-slip movement also may have occurred to the north along the fault between this region and the Kern Mountains.

CONCLUSION

Paleomagnetic data for the Kalamazoo Tuff apparently indicate a 30° clockwise rotation of the Schell Creek Range relative to the Kern Mountains region (Figures 11*b* and 12). This rotation implies differential extention that is most likely accommodated by distributed strike-slip faulting or north-south shortening. Geologic evidence also indicates a north-south variation in the magnitude of extension, where the most highly extended region is roughly coincident with the distribution of the Kalamazoo Tuff. Differential extension between the highly extended region and the less extended region to the north was probably initially accommodated by

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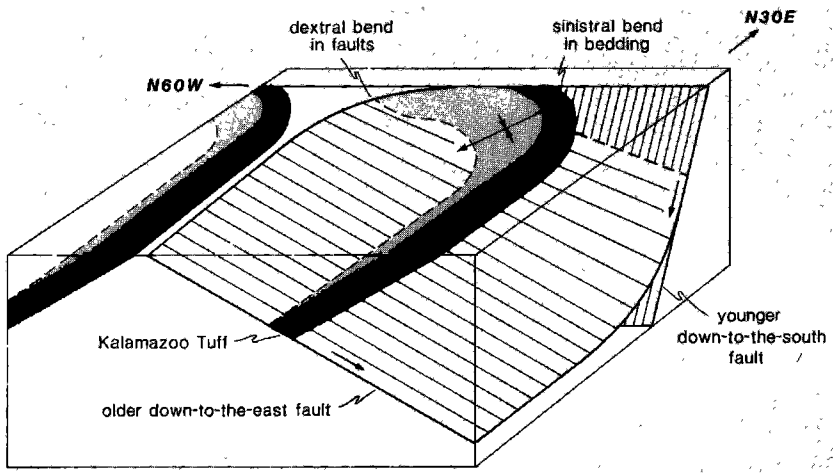


Fig. 13. Schematic block diagram of region along southern flank of the Kern Mountains (Figure 2). Diagram illustrates how apparent oroclinal bends of bedding can be due to drag along a younger down-to-the-south fault instead of vertical axis rotations. Arrows parallel to fault traces on the sides of the block indicate slip directions.

strike-slip faulting along the southern margin of the Kern Mountains. However, large changes in the strike of units near this fault are shown to reflect younger superimposed sets of orthogonal normal faults rather than oroclinal bending associated with strike-slip movement. Differential extension along the southern margin of the highly extended region was apparently accommodated by a zone of complex block faulting and tilting rather than a strike-slip fault.

Along with information on the style of structural deformation in east central Nevada, this study further illustrates a number of problems associated with the paleomagnetic study of ash flow tuffs. Perhaps the most common and most difficult problem to ascertain is the initial variation in the eutaxitic structure caused by emplacement of tuffs over an irregular topography. Although the angular discordance between the Kalamazoo Tuff and underlying strata is generally small, stratigraphic relations beneath this unconformity imply that significant structural and perhaps topographic relief was developed prior to eruption of the tuff [*Gans et al.*, 1989]. Clearly, the first tuff erupted over an irregular topography is most susceptible to the occurrence of topographically related inconsistencies within its eutaxitic structure. For example, the Amalia Tuff of northern New Mexico was emplaced on an irregular surface of intermediate composition volcanic rocks, and the observed tilt-corrected dispersion of its site mean paleomagnetic directions is also high [*Hagstrum and Lipman*, 1986]. The success of *Grommé et al.* [1972] in accurately correlating ash flow tuffs on the basis of paleomagnetic directions was enhanced by their sampling within thick sequences of ash flow units and their use, where available, of the cooling units' planar tops for structural corrections.

Other problems associated with the interpretation of paleomagnetic data for the Kalamazoo Tuff include rheomorphic flow and secular variation of the geomagnetic field during its magnetization. This study presents strong evidence that rheomorphic flow within the Kalamazoo Tuff continued after blocking of its remanent magnetization at temperatures below 575°C. *Rosenbaum* [1986] attributes variations in paleomagnetic directions of a thick welded tuff in southern Nevada to a more subtle postblocking plastic deformation associated with welding. However, the structures associated with viscous flowage [e.g., *Schmincke and Swanson*, 1967] usually can be clearly identified in outcrop exposures, and thinner outflow sheets are probably less susceptible to this type of deformation. Significant secular variation is also shown to have occurred between magnetization of individual tuff units, as well as between magnetization of the vitrophere and devitrified portion of unit B. The effects of secular variation could be minimized by restricting sampling to a distinct portion of a single cooling unit.

Acknowledgments. We thank Sherman Grommé for suggesting this study, for helpful discussions, and for a careful review of the manuscript. Thanks also to Joe Rosenbaum, John Geissman, Jim Diehl, and an anonymous associate editor for their constructive reviews which greatly enhanced the final version of the manuscript. Vicky Pease and Dave Parks assisted with the fieldwork and made most of the laboratory measurements. Material support for PBG was provided by NSF grants EAR 82-06399 (awarded to E. L. Miller and G. A. Mahood) and EAR 84-1876 (awarded to E. L. Miller).

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(Received April 14, 1988;
revised September 26, 1988;
accepted October 12, 1988.)

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